MSG No. 206.38 $Ia\bar{3}1'$ [Type II, cubic]

Table 1: Wyckoff site: 8a, site symmetry: .-3'.1'

No.	position	mapping
1	$[0,\ 0,\ 0]$	[1,5,6,13,17,18,49,53,54,61,65,66]
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	[2,10,11,14,22,23,50,58,59,62,70,71]
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	[3,7,12,15,19,24,51,55,60,63,67,72]
4	$\left[0,\ \frac{1}{2},\ 0\right]$	[4,8,9,16,20,21,52,56,57,64,68,69]
5	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[25,29,30,37,41,42,73,77,78,85,89,90]
6	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[26,34,35,38,46,47,74,82,83,86,94,95]
7	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	[27,31,36,39,43,48,75,79,84,87,91,96]
8	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[28,32,33,40,44,45,76,80,81,88,92,93]

Table 2: Wyckoff site: 8b, site symmetry: .-3'.1'

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	[1,5,6,37,41,42,49,53,54,85,89,90]
2	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	[2,10,11,38,46,47,50,58,59,86,94,95]
3	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	[3,7,12,39,43,48,51,55,60,87,91,96]
4	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	[4,8,9,40,44,45,52,56,57,88,92,93]
5	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	[13,17,18,25,29,30,61,65,66,73,77,78]
6	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	[14,22,23,26,34,35,62,70,71,74,82,83]
7	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	[15,19,24,27,31,36,63,67,72,75,79,84]
8	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	[16,20,21,28,32,33,64,68,69,76,80,81]

Table 3: Wyckoff site: 16c, site symmetry: .3.1'

No.	position	mapping
1	$[x,\ x,\ x]$	[1,5,6,49,53,54]
2	$\left[x,\ -x,\ \frac{1}{2}-x\right]$	[2,10,11,50,58,59]
3	$\left[\frac{1}{2}-x,\ x,\ -x\right]$	[3,7,12,51,55,60]
4	$\left[-x,\ \frac{1}{2}-x,\ x\right]$	[4,8,9,52,56,57]
5	$[-x,\ -x,\ -x]$	[13,17,18,61,65,66]
6	$\left[-x,\ x,\ x+\frac{1}{2}\right]$	[14,22,23,62,70,71]
7	$\left[x+\frac{1}{2},\ -x,\ x\right]$	[15,19,24,63,67,72]
8	$\left[x,\ x+\frac{1}{2},\ -x\right]$	[16,20,21,64,68,69]
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[25,29,30,73,77,78]
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -x\right]$	[26,34,35,74,82,83]
11	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	[27,31,36,75,79,84]
12	$\left[\frac{1}{2}-x,\ -x,\ x+\frac{1}{2}\right]$	[28,32,33,76,80,81]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[37,41,42,85,89,90]
14	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x\right]$	[38,46,47,86,94,95]
15	$\left[x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	[39,43,48,87,91,96]
16	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-x\right]$	[40,44,45,88,92,93]

Table 4: Wyckoff site: `24d`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,2,49,50]`
2	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	`[3,28,51,76]`
3	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[4,27,52,75]`
4	$\left[\frac{1}{4},\ x,\ 0\right]$	`[5,12,53,60]`
5	$\left[0,\ \frac{1}{4},\ x\right]$	`[6,9,54,57]`
6	$\left[\frac{1}{2},\ \frac{1}{4},\ -x\right]$	`[7,35,55,83]`
7	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ 0\right]$	`[8,34,56,82]`
8	$\left[\frac{1}{4},\ -x,\ \frac{1}{2}\right]$	`[10,32,58,80]`
9	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-x\right]$	`[11,31,59,79]`
10	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[13,14,61,62]`
11	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[15,40,63,88]`
12	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[16,39,64,87]`
13	$\left[\frac{3}{4},\ -x,\ 0\right]$	`[17,24,65,72]`
14	$\left[0,\ \frac{3}{4},\ -x\right]$	`[18,21,66,69]`
15	$\left[\frac{1}{2},\ \frac{3}{4},\ x\right]$	`[19,47,67,95]`
16	$\left[\frac{1}{4},\ x+\frac{1}{2},\ 0\right]$	`[20,46,68,94]`
17	$\left[\frac{3}{4},\ x,\ \frac{1}{2}\right]$	`[22,44,70,92]`
18	$\left[0,\ \frac{1}{4},\ x+\frac{1}{2}\right]$	`[23,43,71,91]`
19	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[25,26,73,74]`
20	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[29,36,77,84]`
21	$\left[\frac{1}{2},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	`[30,33,78,81]`
22	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[37,38,85,86]`
23	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[41,48,89,96]`
24	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	`[42,45,90,93]`

Table 5: Wyckoff site: `48e`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,49]`
2	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	`[2,50]`
3	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	`[3,51]`
4	$\left[-x,\ \frac{1}{2}-y,\ z\right]$	`[4,52]`
5	$[z,\ x,\ y]$	`[5,53]`
6	$[y,\ z,\ x]$	`[6,54]`
7	$\left[\frac{1}{2}-y,\ z,\ -x\right]$	`[7,55]`
8	$\left[-z,\ \frac{1}{2}-x,\ y\right]$	`[8,56]`
9	$\left[-y,\ \frac{1}{2}-z,\ x\right]$	`[9,57]`
10	$\left[z,\ -x,\ \frac{1}{2}-y\right]$	`[10,58]`
11	$\left[y,\ -z,\ \frac{1}{2}-x\right]$	`[11,59]`
12	$\left[\frac{1}{2}-z,\ x,\ -y\right]$	`[12,60]`
13	$[-x,\ -y,\ -z]$	`[13,61]`
14	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	`[14,62]`
15	$\left[x+\frac{1}{2},\ -y,\ z\right]$	`[15,63]`
16	$\left[x,\ y+\frac{1}{2},\ -z\right]$	`[16,64]`
17	$[-z,\ -x,\ -y]$	`[17,65]`

continued ...

Table 5

No.	position	mapping
18	$[-y,\ -z,\ -x]$	[18,66]
19	$\left[y+\frac{1}{2},\ -z,\ x\right]$	[19,67]
20	$\left[z,\ x+\frac{1}{2},\ -y\right]$	[20,68]
21	$\left[y,\ z+\frac{1}{2},\ -x\right]$	[21,69]
22	$\left[-z,\ x,\ y+\frac{1}{2}\right]$	[22,70]
23	$\left[-y,\ z,\ x+\frac{1}{2}\right]$	[23,71]
24	$\left[z+\frac{1}{2},\ -x,\ y\right]$	[24,72]
25	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[25,73]
26	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[26,74]
27	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[27,75]
28	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[28,76]
29	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	[29,77]
30	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[30,78]
31	$\left[-y,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$	[31,79]
32	$\left[\frac{1}{2}-z,\ -x,\ y+\frac{1}{2}\right]$	[32,80]
33	$\left[\frac{1}{2}-y,\ -z,\ x+\frac{1}{2}\right]$	[33,81]
34	$\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ -y\right]$	[34,82]
35	$\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ -x\right]$	[35,83]
36	$\left[-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	[36,84]
37	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[37,85]
38	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[38,86]
39	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[39,87]
40	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	[40,88]
41	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	[41,89]
42	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[42,90]
43	$\left[y,\ \frac{1}{2}-z,\ x+\frac{1}{2}\right]$	[43,91]
44	$\left[z+\frac{1}{2},\ x,\ \frac{1}{2}-y\right]$	[44,92]
45	$\left[y+\frac{1}{2},\ z,\ \frac{1}{2}-x\right]$	[45,93]
46	$\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ y\right]$	[46,94]
47	$\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ x\right]$	[47,95]
48	$\left[z,\ \frac{1}{2}-x,\ y+\frac{1}{2}\right]$	[48,96]